

CHAPTER-1

INTRODUCTION

1.1 Introduction

Smart Lit: The Future of Adaptive Street Lighting:

Imagine streets that adjust their lighting in real-time to conserve energy, enhance safety, and promote sustainability. Welcome to Smart Lit, the future of adaptive street lighting. This innovative system leverages advanced technologies like IoT sensors, AI, and energy-efficient LED lighting to create a smarter, more responsive urban infrastructure. Smart Lit represents a significant step towards creating intelligent, adaptive urban infrastructure. By harnessing the power of technology, we can build safer, more sustainable cities that benefit both residents and the environment.

1.2 KEY FEATURES

- 1. Adaptive Lighting:** Adjusts brightness based on ambient light, pedestrian traffic, and weather conditions.
- 2. Real-time Monitoring:** Tracks energy consumption, detects faults, and optimizes performance.
- 3. Energy Efficiency:** Reduces energy waste and lowers carbon emissions.
- 4. Enhanced Safety:** Improves visibility and reduces crime rates.

1.3 BENEFITS

- 1. Sustainable Cities:** Supports eco-friendly urban development.
- 2. Cost Savings:** Reduces energy costs and maintenance expenses.
- 3. Improved Quality of Life:** Enhances public safety and ambiance.

1.4 EXISTING SYSTEM

The existing system of streetlights typically consists of a network of traditional streetlights that are connected to the power grid and operated by a central control system. The streetlights are usually turned on and off manually or using a timer, with fixed brightness levels that are not adjusted based on ambient light levels or other factors. The traditional streetlight system has several limitations, including:

1. **High energy consumption:** Traditional streetlights consume a significant amount of energy, leading to high operational costs and carbon emissions.
 2. **Limited control:** The brightness levels of traditional streetlights are fixed, and there is limited control over their operation and maintenance.
 3. **High maintenance costs:** Traditional streetlights require regular maintenance and replacement of bulbs, leading to high maintenance costs and downtime.
 4. **Poor visibility:** Traditional streetlights may not provide adequate visibility for drivers and pedestrians, leading to safety risks.
 5. **Lack of connectivity:** Traditional streetlights are not connected to the internet or other wireless networks, limiting their potential to collect and share data.
- Overall, the existing system of streetlights is inefficient and outdated, and there is a need for more advanced and sustainable solutions such as smart streetlights.

1.5 PROPOSED SYSTEM

A proposed system of smart streetlights involves the use of advanced technologies such as sensors, wireless communication, and data analytics to create a more efficient, sustainable, and safe street lighting system. Here are some of the key features and components of a smart streetlight system:

- 1. LED lights:** LED lights are highly energy-efficient and have a longer lifespan 15 than traditional bulbs, reducing maintenance costs and carbon emissions.
- 2. Light sensors:** Light sensors such as LDR (Light-Dependent Resistor) sensors can detect the ambient light levels and adjust the brightness of the streetlights accordingly, saving energy and enhancing visibility.
- 3. Motion sensors:** Motion sensors such as PIR (Passive Infrared) sensors can detect the presence of pedestrians and vehicles and adjust the brightness of the streetlights, improving safety and reducing energy consumption.

1.5 BLOCK DIAGRAM

Figure 1.1 shows the internal operation of the prototype.

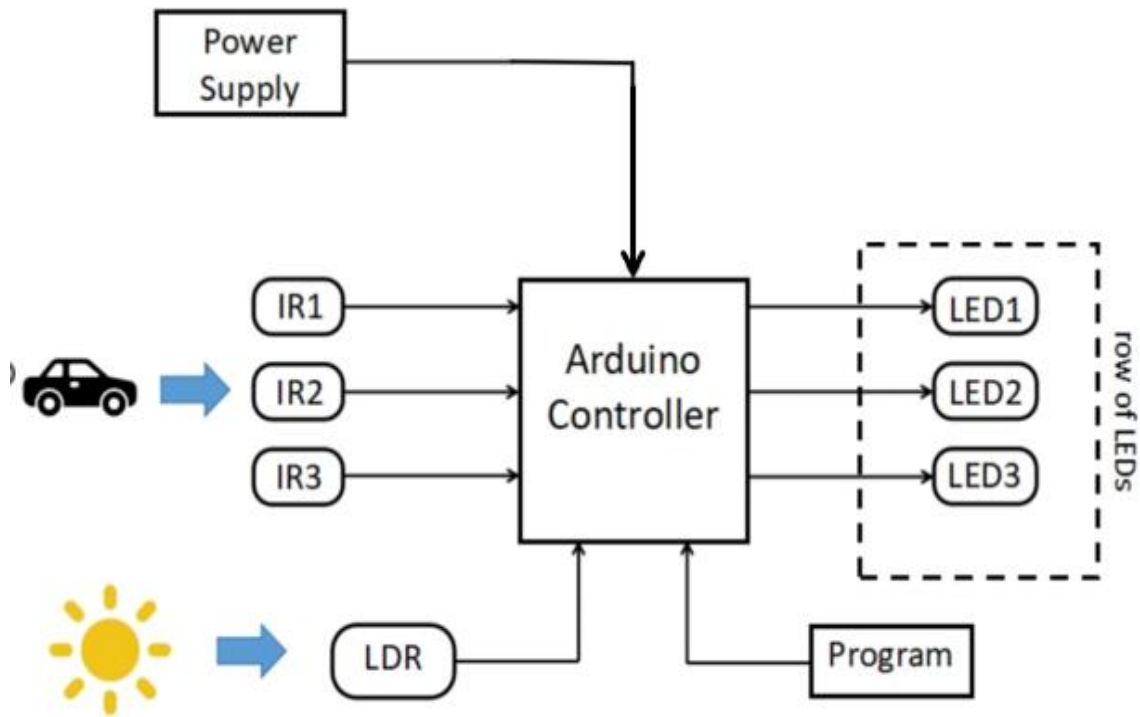


Figure 1.1 SMART LIT : The future of Adaptive Street lighting

- **ARDUINO:**

Arduino boards are open-source, programmable electronics platforms used for creating interactive physical objects and devices. They offer a versatile and easy-to-use way to prototype and build projects involving sensors, actuators, and other electronic components. Key features include a microcontroller, input/output pins, a USB interface, and a programming environment. .

- **LDR (Light Dependent Resistor):**

The LDR is a type of resistor that changes its resistance based on the amount of light that it is exposed to. In the Laser Guardian system, the LDR is used to detect changes in the laser beam's intensity, indicating an intrusion.

- **LED (Light Emitting Diode):**

The LED is used to provide visual feedback to the user, indicating that the system is turned on and functioning properly.

- **Power Supply:**

The power supply provides the necessary electrical power to the Laser Guardian system.

- **9V BATTERY:**

A 9V battery is a small, rectangular battery that provides 9 volts of electrical power. It's often used in small electronics and projects.

- **IR SENSOR:**

An IR sensor (Infrared sensor) is an electronic device that detects infrared light (IR) from objects in its surroundings. It is commonly used to detect motion, distance, or presence of an object

Together, these components work to create a reliable and effective project for detecting and working of this project *SMART LIT : The future of Adaptive Street lighting*

1.7 DESIGN METHODOLOGY

For this project, there are many criteria that can be added and optimized in order to increase the effectiveness of the system. The workflow of proposed system is shown below

- The Arduino is used to send the signal to the LED and take the signal from the IR sensor and for this, we dump the program into it. By this when there is any vehicle movement the IR sensor sends the signal to the Arduino by which the light glows more brighter.
- IR sensors can be used to detect movement in a defined angle range.

- The 9v battery is designed to save the power so we use a 9v battery as a power supply, to consume low power and give us the output.
- LDR sensor for the light detection ,it detects the light and makes the street light glow .
- Light-emitting diodes (LEDs) are semiconductor devices that emit light when an electric current passes through them. They are used in many applications, including TV backlighting and general illumination.

The flow chart of the project is as shown below in **Figure 1.2**

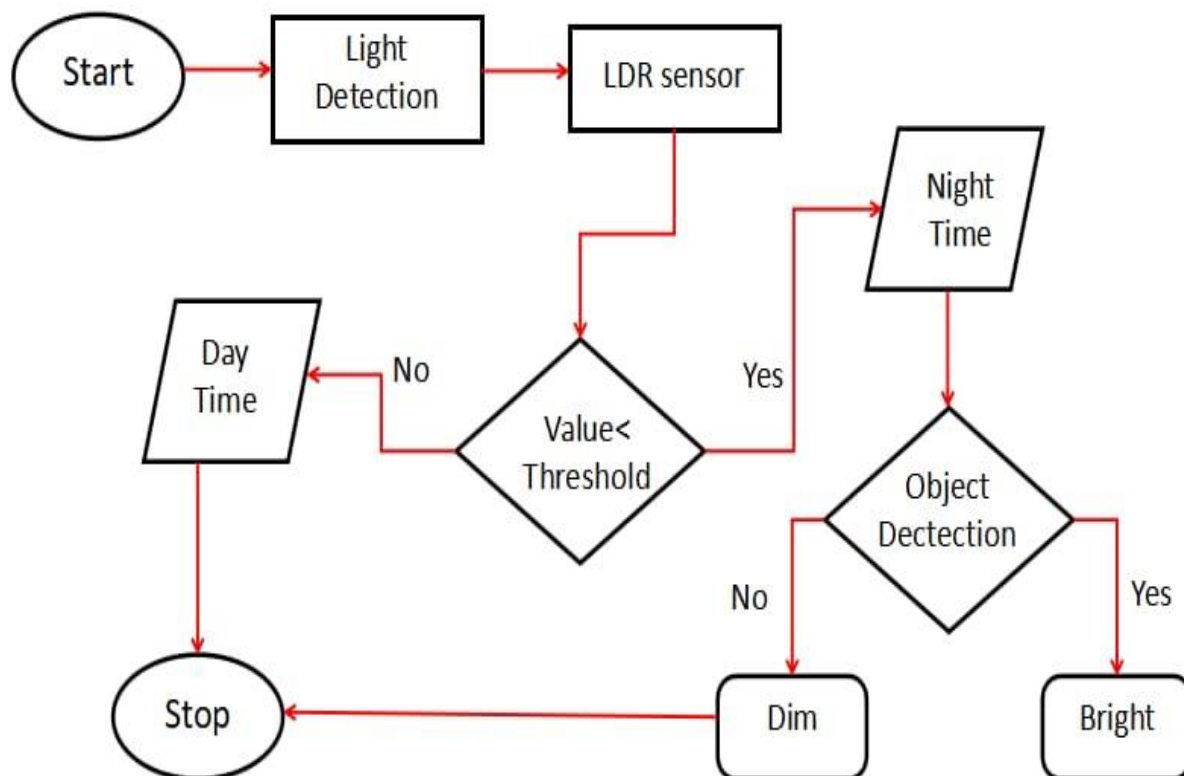


Figure 1.2 Design flow of the system

CHAPTER 2

HARDWARE COMPONENTS

2.1 ARDUINO

2.1.1 Introduction:

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IOT applications, wearable, 3D printing, and embedded environments.

2.1.2 Designing of a Arduino:

In recent times, the UNO board has become synonymous with Arduino.



Figure 2.1 Components of a typical Arduino

Components of Arduino UNO board

The major components of Arduino UNO board are as follows:

1. USB connector
2. Power port
3. Microcontroller
4. Analog input pins
5. Digital pins
6. Reset switch
7. Crystal oscillator
8. USB interface chip
9. TX RX LEDs

Now let's take a closer look at each component.

2.1.3 Theory of Operation:

1. USB connector:

This is a printer USB port used to load a program from the Arduino IDE onto the Arduino board. The board can also be powered through this port.

2. Power port:

The Arduino board can be powered through an AC-to-DC adapter or a battery. The power source can be connected by plugging in a 2.1mm center-positive plug into the power jack of the board.

2.1mm center-positive plug

The Arduino UNO board operates at a voltage of 5 volts, but it can withstand a maximum voltage of 20 volts. If the board is supplied with a higher voltage, there is a voltage regulator (it sits between the power port and USB connector) that protects the board from burning out.

3. Microcontroller:

Atmega328P microcontroller

It is the most prominent black rectangular chip with 28 pins. Think of it as the brains of your Arduino. The microcontroller used on the UNO board is Atmega328P by Atmel (a major microcontroller manufacturer). Atmega328P has the following components in it:

- **Flash memory** of 32KB. The program loaded from Arduino IDE is stored here.
- **RAM** of 2KB. This is a runtime memory.
- **CPU:** It controls everything that goes on within the device. It fetches the program instructions from flash memory and runs them with the help of RAM.
- **Electrically Erasable Programmable Read Only Memory (EEPROM)** of 1KB. This is a type of nonvolatile memory, and it keeps the data even after device restart and reset.

Atmega328P is pre-programmed with bootloader. This allows you to directly upload a new Arduino program into the device, without using any external hardware programmer, making the Arduino UNO board easy to use.

4. Analog input pins:

The Arduino UNO board has 6 analog input pins, labelled “Analog 0 to 5.” These pins can read the signal from an analog sensor like a temperature sensor and convert it into a digital value so that the system understands. These pins just measure voltage and not the current because they have very high internal resistance. Hence, only a small amount of current flows through these pins. Although these pins are labeled analog and are analog input by default, these pins can also be used for digital input or output.

5. Digital pins:

You can find these pins labeled “Digital 0 to 13.” These pins can be used as either input or output pins. When used as output, these pins act as a power supply source for the components connected to them. When used as input pins, they read the signals from the component connected to them. When digital pins are used as output pins, they supply 40 milliamps of current at 5 volts, which is more than enough to light an LED.

Some of the digital pins are labeled with tilde (~) symbol next to the pin numbers (pin numbers 3, 5, 6, 9, 10, and 11). These pins act as normal digital pins but can also be used for Pulse-Width Modulation (PWM), which simulates analog output like fading an LED in and out.

6. Reset switch:

When this switch is clicked, it sends a logical pulse to the reset pin of the Microcontroller, and now runs the program again from the start. This can be very useful if your code doesn’t repeat, but you want to test it multiple times.

7. Crystal oscillator:

Crystal oscillator

This is a quartz crystal oscillator which ticks 16 million times a second. On each tick, the microcontroller performs one operation, for example, addition, subtraction, etc.

8. USB interface chip:

USB interface chip

Think of this as a signal translator. It converts signals in the USB level to a level that an Arduino UNO board understands.

9. TX – RX LEDs:

TX – RX indicator

TX stands for transmit, and RX for receive. These are indicator LEDs which blink whenever the UNO board is transmitting or receiving data.

Now that you have explored the Arduino UNO board, you have started your journey toward building your first IoT prototype. In the next article, we will discuss Arduino programming and do a few experiments with Arduino and LEDs.

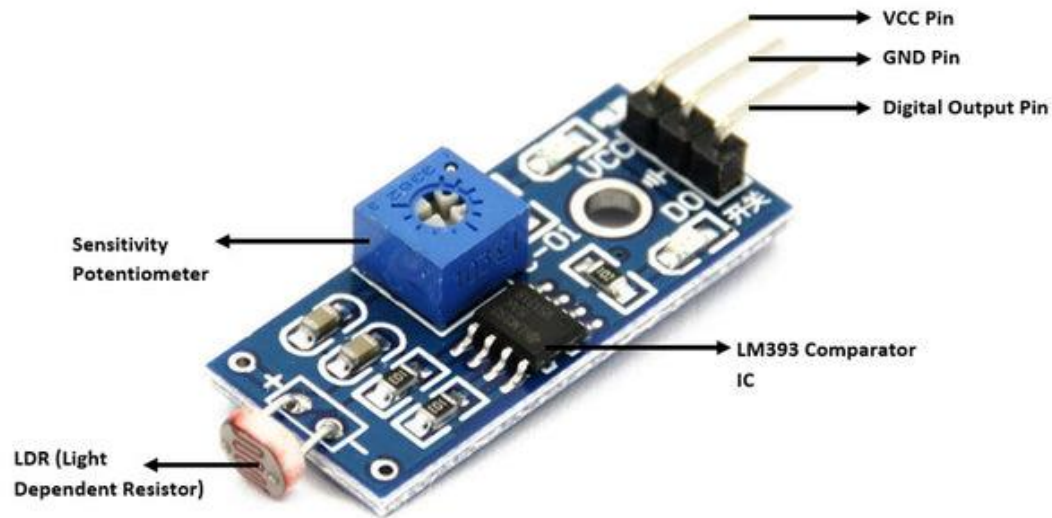


Figure 2.2 LDR

2.2 LDR (Light Dependent Resistor)

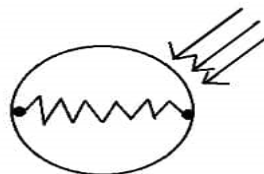
2.2.1 Introduction:

LDR (Light Dependent Resistor) as the name states is a special type of resistor that works on the photoconductivity principle means that resistance changes according to the intensity of light. Its resistance decreases with an increase in the intensity of light.

It is often used as a light sensor, light meter, Automatic street light, and in areas where we need to have light sensitivity. LDR is also known as a Light Sensor. LDR are usually available in 5mm, 8mm, 12mm, and 25mm dimensions.

2.2.2 Structure of an LDR

The Light-dependent resistors made with photosensitive semiconductor materials like Cadmium Sulphides (CdS), lead sulfide, lead selenide, indium antimonide, or cadmium selenide and they are placed in a Zig-Zag shape as shown in **Figure 2.2**.



Two metal contacts are placed on both ends of the Zig-Zag shape these metal contacts help in creating a connection with the LDRs.

Now, a transparent coating is applied on the top so that the zig-zag-shaped photosensitive material gets protected and as the coating is transparent the LDR will be able to capture light from the outer environment for its working.

2.2.3 LDR Working Principle:

It works on the principle of photoconductivity whenever the light falls on its photoconductive material, it absorbs its energy and the electrons of that photoconductive material in the valence band get excited and go to the conduction band and thus increasing the conductivity as per the increase in light intensity.

Also, the energy in incident light should be greater than the bandgap energy so that the electrons from the valence band get excited and go to the conduction band.

The LDR has the highest resistance in dark around 10^{12} Ohm and this resistance decreases with the increase in Light.

2.2.4 LDR Applications

The LDR applications is as follows

- The photoresistor is generally used in detecting the presence and intensity of light
- Used in automatic lights that switch on and off according to light
- Simple Smoke Detector Alarm, Clock with automatic light
- Optical circuit design
- Photo proximity switch
- Laser-based security systems
- Solar Street Lamps
- Camera light meters
- Can be used in Dynamic Compressors, some compressors use LDR and LED connected to the signal source to create changes in signal gain.

2.2.5 Limitations

The limitations of LDR are as follows

- LDRs require a few milliseconds or more to respond fully to the changes in light intensity, i.e. they require a few seconds to return to their normal resistances once the light source is removed.
- The sensitivity of a Light-dependent resistor varies with the light wavelength. If the wavelength is outside a certain range, it will not affect the resistance at all.
- Light-dependent resistors have lower sensitivity than photodiodes and phototransistors.

2.3 LED (Light Emitting Diode)

2.3.1 Introduction:

Light-emitting diode (LED) is a widely used standard source of light in electrical equipment as shown in **Figure 2.3**. It has a wide range of applications ranging from your mobile phone to large advertising billboards. They mostly find applications in devices that show the time and display different types of data.

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction.

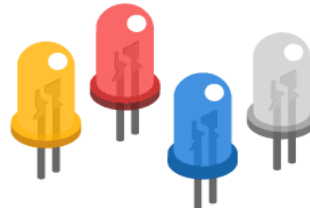


Figure 2.3 LED

Light-emitting diodes are heavily doped p-n junctions. Based on the semiconductor material used and the amount of doping, an LED will emit colored light at a particular spectral wavelength when forward biased. As shown in the figure, an LED is encapsulated with a transparent cover so that emitted light can come out.

2.3.2 LED Symbol

The LED symbol as shown in **Figure 2.4** is the standard symbol for a diode, with the

addition of two small arrow denoting the emission of light.

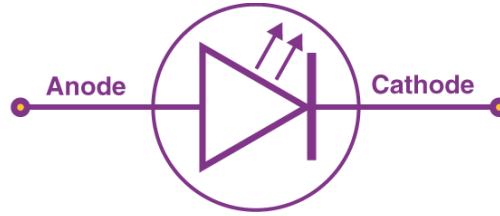


Figure 2.4 LED symbol

2.3.3 LED working principle

When the diode is forward biased, the minority electrons are sent from $p \rightarrow n$ while the minority holes are sent from $n \rightarrow p$ as shown in figure 2.5. At the junction boundary, the concentration of minority carriers increases. The excess minority carriers at the junction recombine with the majority charges carriers.

The energy is released in the form of photons on recombination. In standard diodes, the energy is released in the form of heat. But in light-emitting diodes, the energy is released in the form of photons. We call this phenomenon electroluminescence. Electroluminescence is an optical phenomenon, and electrical phenomenon where a material emits light in response to an electric current passed through it. As the forward voltage increases, the intensity of the light increases and reaches a maximum.

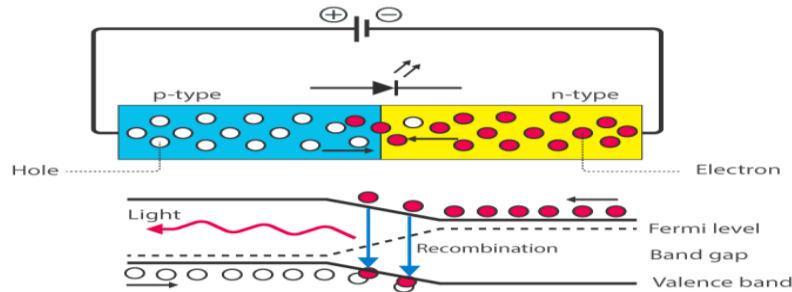


Figure 2.5 Working of an LED

2.3.4 Uses of LED

LEDs find applications in various fields, including optical communication, alarm and security systems, remote-controlled operations, robotics, etc. It finds usage in many areas because of its long-lasting capability, low power requirements, swift response time, and fast switching capabilities. Below are a few standards LED uses:

- Used for TV back-lighting
- Used in displays

- Used in Automotives
- LEDs used in the dimming of lights



Figure 2.6 IR Sensor

2.4 IR Sensor

2.4.1 Introduction:

IR Sensor

An IR sensor is an electronic device that detects IR radiation falling on it. [Proximity sensors](#) (used in touchscreen phones and edge-avoiding robots), contrast sensors (used in line following robots), and obstruction counters/sensors (used for counting goods and in burglar alarms) are some applications involving IR sensors.

2.4.2 IR Sensor Working Principle

An IR sensor consists of two parts: the emitter circuit and the receiver circuit. This is collectively known as a photo-coupler or an optocoupler.

The emitter is an IR LED, and the detector is an IR photodiode. The IR photodiode is sensitive to the IR light emitted by an IR LED. The photodiode's resistance and output voltage change in proportion to the IR light received. This is the underlying working principle of the IR sensor.

The type of incidence can be direct incidence or indirect incidence. In direct incidence, the IR, LED is placed in front of a photodiode with no obstacle.

In indirect incidence, both diodes are placed side by side with an opaque object in front of the sensor. The light from the IR LED hits the opaque surface and reflects back to the photodiode.

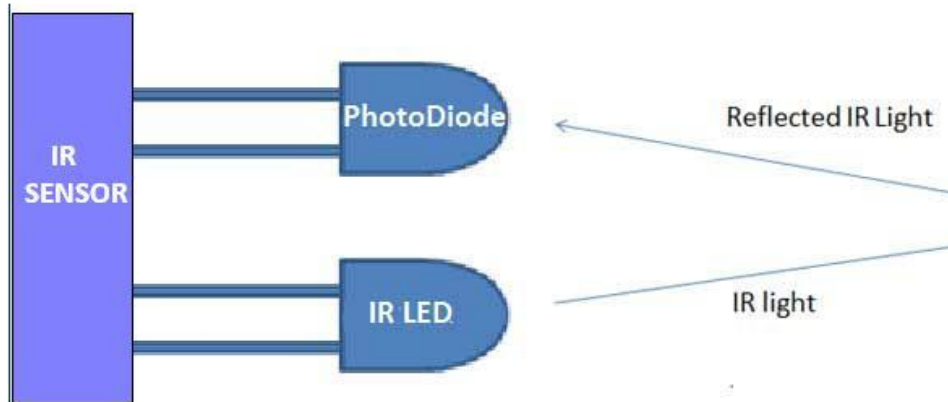


Figure 2.7 Working of IR Sensor

- **IR Sensor Working Principle**

IR Receiver LED and IR Transmitter LED

An IR receiver LED and an IR transmitter LED are both types of light-emitting diodes (LEDs) that are used in infrared (IR) communication.

An IR receiver LED is a device that detects infrared signals from remote controls and other IR sources. It is typically a small, clear, or translucent device that is sensitive to IR light in a specific frequency range.

When an IR signal is detected, the IR receiver LED will emit a small amount of visible light, which can be used to confirm that a signal has been received.

An IR transmitter LED, on the other hand, is a device that emits infrared light in order to send signals to other devices. It is typically a small, clear, or translucent device that emits IR light in a specific frequency range. IR transmitter LEDs are commonly used in remote controls and other IR signaling devices.

In summary, the IR receiver LED detects the IR signal from the remote, and the IR transmitter LED emits the IR signal from the remote.

IR Sensor Module

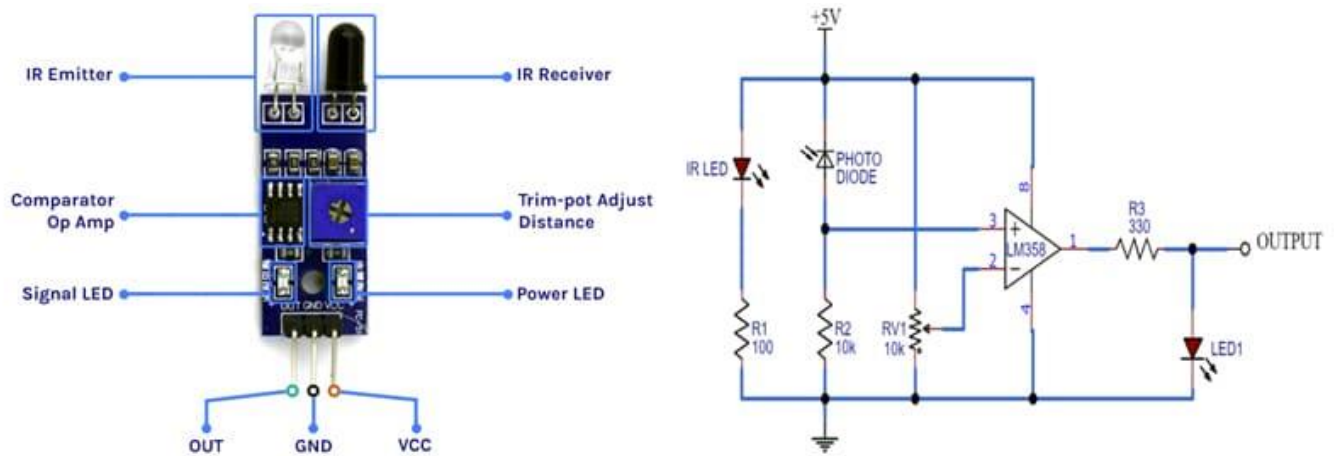


Figure 2.8 Parts of IR Sensor

IR Sensor Module with Circuit

An IR sensor module is a device that contains an IR receiver LED and other components that are used to detect and process IR signals. It typically includes an IR receiver LED, a signal amplifier, and a demodulator circuit.

The IR receiver LED is used to detect IR signals, while the signal amplifier and demodulator circuit are used to amplify and process the received signal, respectively.

IR sensor modules are widely used in various electronic applications such as remote control, motion detection, proximity sensing, and more. They are commonly used in consumer electronics, robotics, and automation systems.

IR sensor modules come in various forms, such as simple IR receiver modules and complex IR sensor modules with additional features such as [signal processing](#) and signal filtering. Some IR sensor modules also provide an output in a digital format that can be read by a microcontroller or microprocessor.

IR sensors find a wide variety of applications in various fields. Let's take a look at a few of them

2.5 POWER SUPPLY

2.5.1 Introduction:

Power supply is a supply of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply unit or PSU. A power supply may include a power distribution system as well as primary or secondary sources of energy such as conversion of one form of electrical power to another desired form and voltage, typically involving converting AC line voltage to a well-regulated lower-voltage DC for electronic devices. Low voltage, low power DC power supply units are commonly integrated with the devices they supply, such as computers and household electronics.

- Batteries.
- Chemical fuel cells and other forms of energy storage systems.
- Solar power.
- Generators or alternators.

2.5.2 Block Diagram: The below **Figure 2.9** shows of flow of current in the form of a block diagram

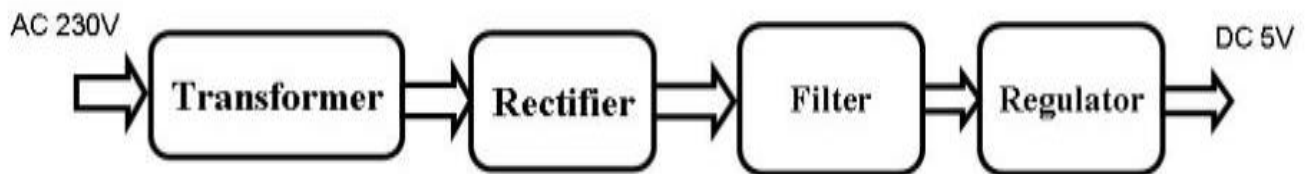


Figure 2.9 Block diagram for the flow of current

2.6 9V BATTERY

A battery is a type of linear power supply that offers benefits that traditional line operated power supplies lack: mobility, portability and reliability. A battery consists of multiple electrochemical cells connected to provide the voltage desired. **Figure 2.10** shows Hi-Watt 9V battery.



Figure 2.10 Hi-Watt 9V Battery

A 9V battery comes in two basic types. These are primary 9V batteries and 9-volt rechargeable batteries. The primary ones have different chemistries or compositions. Rechargeable batteries are also made of various chemical compositions with slight differences in some features. Primary cells are either alkaline batteries, Lithium, zinc-air, Zinc chloride, silver oxide, and other variants. On the other hand, rechargeable 9-volt batteries are made up of such chemistries as rechargeable alkaline batteries, Lithium-ion, Lithium Iron Phosphate (LiFePO₄), Lithium Manganese Oxide (LiMn₂O₄), Lithium Titanate (Li₂TiO₃), Lithium Nickel Manganese Cobalt Oxide (LiNiMnCoO₂), Lithium Cobalt Oxide (LiCoO₂), Nickel Cadmium (NiCad), Nickel Metal Hydride (NiMH), Lead Acid. Different chemistries have different characteristics. This affects the performance and compatibility with your chosen devices.

Figure 2.11 Parts of 9V Battery



2.7 Breadboard

2.7.1 Introduction:

A breadboard (sometimes called protoboard) is essentially the foundation to construct and prototype electronics. A breadboard allows for easy and quick creation of temporary electronic circuits or to carry out experiments with circuit design. Breadboards enable developers to easily connect components or wires thanks to the rows and columns of internally connected spring clips underneath the perforated plastic enclosure. The grid is made up of perfectly aligned spring clip holes that are 0.1". A breadboard can be divided into two segments, which are called the bus strip and the terminal strip as shown in below **Figure 2.12..**

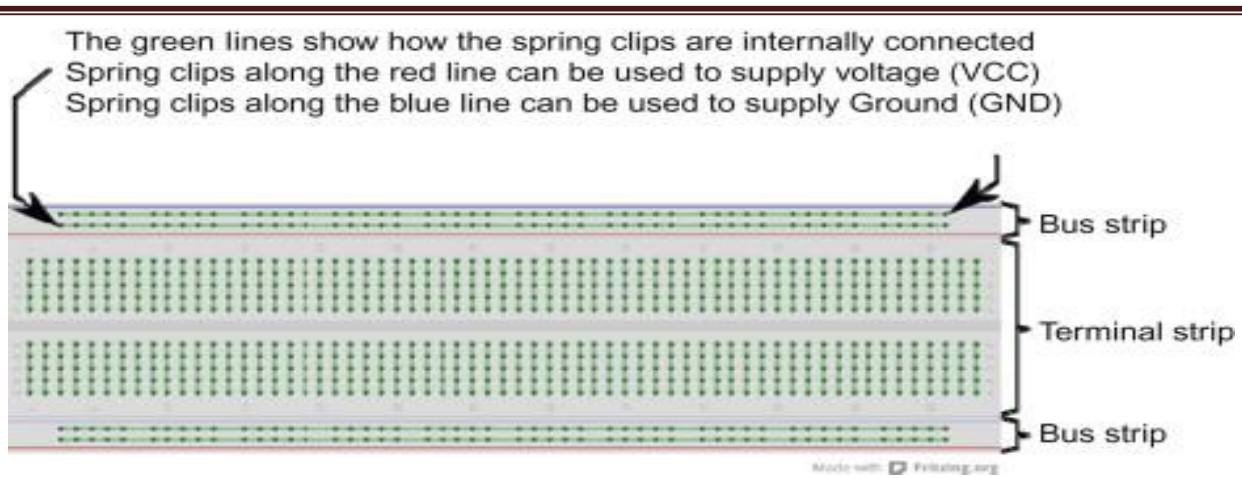


Figure 2.12 Breadboard

The bus strip consists of two long lines of spring clips running across the board, these lines are used to provide supply voltage (VCC) and ground (GND) to the circuit. Typically, the supply voltage line is marked in red and the ground line is marked in blue. All spring clips along each line are internally connected, making both supply voltage and ground signals conveniently accessible from any part of the breadboard.

2.7.2 VCC

VCC is the supply voltage that the microcontroller system uses. The Launchpad can run properly when a supply voltage of 1.8–3.6 V is available. In most cases, the VCC of our MSP430-based projects will be ~3 V. We will use this terminology and referencing this supply voltage throughout the book.

2.7.3 Ground

All electrical current in a system needs to flow back to ground. In a typical microcontroller project, all components flow back to Ground, which is why this is usually referred to as “common ground.” At GND, this is usually a 0-V reading. We will touch on VCC and GND throughout the book.

The terminal strip is the main area, it is usually separated into two sides by a notch that runs along the middle of the board. Each side has many lines that are made up of five internally connected spring clip holes. The five spring clips on each line of the terminal strip are connected internally, allowing for component connections.

2.8 Connecting Wires

2.8.1 Introduction:

A connecting wire allows travels the electric current from one point to another point without resistivity. Resistance of connecting wire should always be near zero. Copper wires have low resistance and are therefore suitable for low resistance. **Wire** as defined by "Compton's Encyclopaedia" is a strand of metal in the form of a flexible thread or slender rod. This is the most common form of conductors for power transmission, smaller electrical signals, or as resistors. Insulated wires are most often used in electronic circuits. The insulation provides electrical insulation and mechanical and chemical protection.

2.8.2 Different Types of Connecting Wires

Wires are either solid or stranded. Most wires are round, occasionally square or rectangular conductors are used, such as integrated circuit external leads. Metals usually used in making wires are aluminium, alloy and copper. Insulation is made up of rubber or non-conductive materials and can come in different sizes and colours.

- **Solid Wires**

These wires are single solid wires with rubber insulation. Usually used in connecting circuits or wiring connections in a protoboard as shown in **figure 2.13**.



Figure 2.13 Solid Wires

- **Standard Wire**

Is a group of wires used as a single wire. Due to their high flexibility, stranded wires are the most commonly used conductors. There is no sharp distinction between stranded wire and cable wires. Stranded constructions vary in the size, number and configuration of the individual strands as shown in **Figure 2.14**. All stranded designation systems relate to the total cross-sectional area of the conductor. The cross-sectional area of metal determines the electrical resistance and current carrying capacity of the conductor and is important for the proper size selection for a specific application.

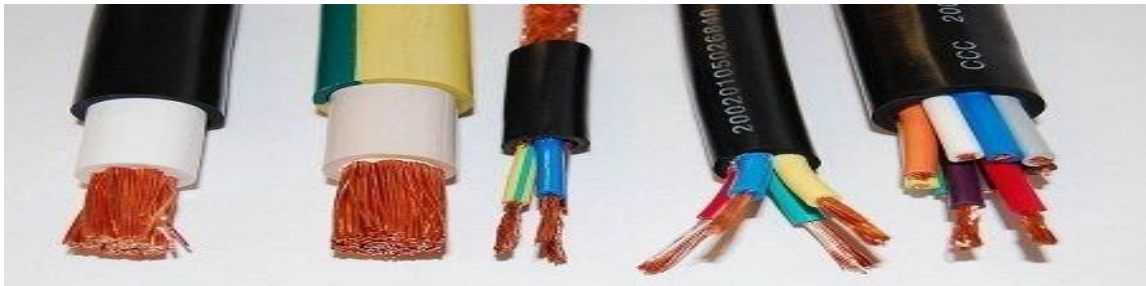


Figure 2.14 Standard Wires

2.8.3 Jumper Wires

Jumper wires typically come in three versions: male-to-male, male-to-female and female-to-female. The difference between each is in the end point of the wire. Male ends have a pin protruding and can plug into things, while female ends do not and are used to plug things into. Male-to-male jumper wires are the most common and what you likely will use most often. When connecting two ports on a breadboard, a male-to-male wire is what you'll need as shown in

Figure 2.15.

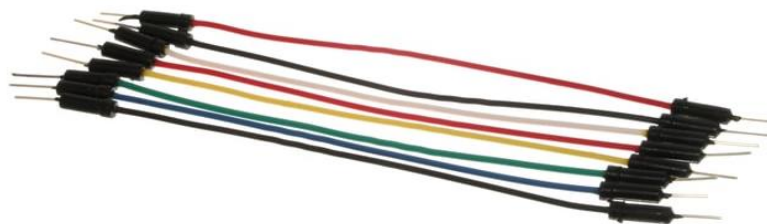


Figure 2.15 Jumper Wires

3. RESULTS AND ANALYSIS

Here is what we did, we have created system consisting of a microcontroller i.e., Arduino Uno which is programmed in such a way that it controls the streetlights based on environmental factors such as Human Activity detection, Object detection and Light intensity of the environment. This system contains a PIR sensor which detects the motion activity in the environment PIR sensors are designed to detect changes in the infrared radiation emitted by objects in their field of view. They consist of a pyroelectric sensor that generates an electrical signal when it detects infrared radiation. LDR sensor is connected to system which detects the light intensity of the environment. When light falls on the LDR sensor, the photons in the light excite the electrons in the sensor material, causing them to move from the valence band to the conduction band. This movement of electrons reduces the resistance of the sensor. The greater the intensity of light, the more electrons are excited, and the lower the resistance of the sensor.

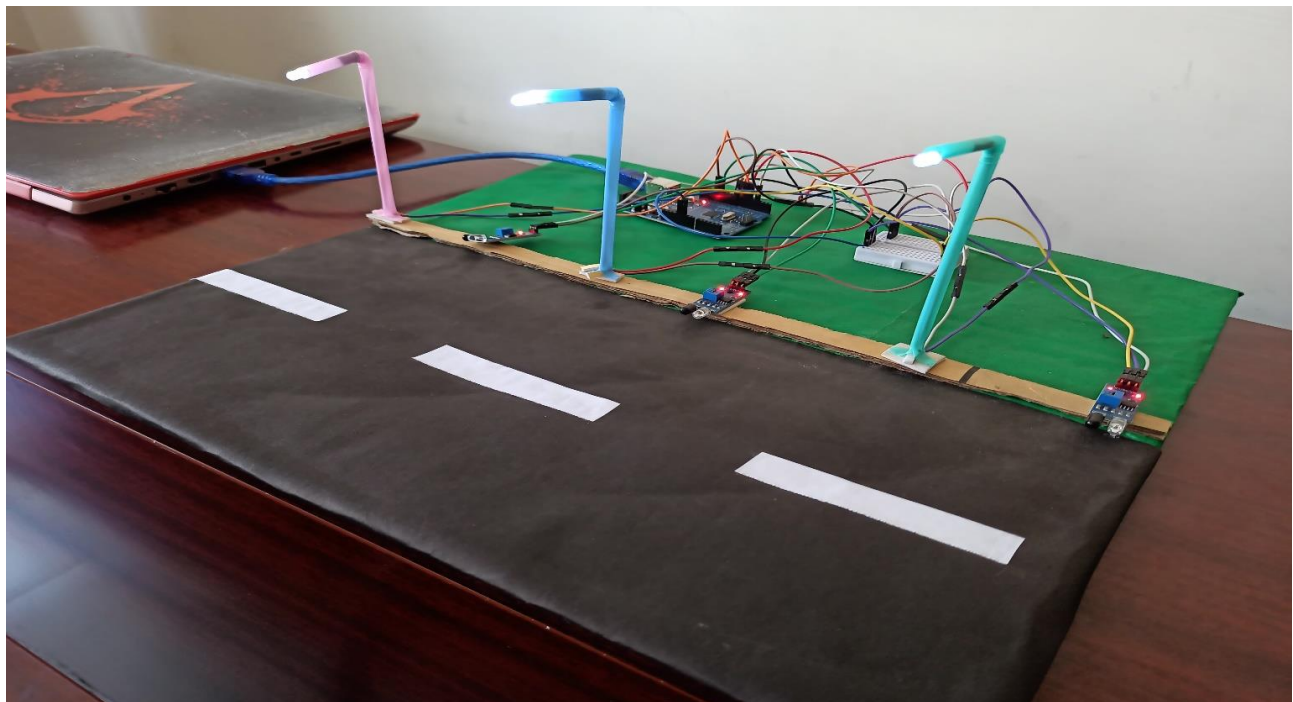


Figure 3.1 Project outlet

4. CONCLUSION

In conclusion, *Smart Lit: The Future of Adaptive Street Lighting* based on IR sensor and LDR sensor can offer several benefits. The IR sensor can detect the presence of people or vehicles, and the LDR sensor can measure the ambient light levels. By combining the two sensors, the system can automatically adjust the brightness of the streetlights based on the surrounding environment and the presence of people or vehicles. This system can lead to significant energy savings, as the streetlights will only operate at full brightness when necessary. It can also reduce light pollution and improve safety by providing better lighting when people or vehicles are present. However, the implementation of such a system would require careful planning and consideration of factors such as sensor placement, communication protocols, and power management. Additionally, maintenance and repair of the system would need to be considered to ensure reliable operation. Overall, a smart streetlight system based on IR sensor and LDR sensor has the potential to be a cost effective and sustainable solution for public lighting

5. REFERENCES

1. "Smart Street Lighting: A Review of the Current State of the Art" (Journal of Lighting Research and Technology)
2. "Adaptive Street Lighting: A Survey of Technologies and Applications" (IEEE Transactions on Intelligent Transportation Systems)
3. "Energy Efficiency in Street Lighting: A Comparative Study of Different Technologies" (Energy and Buildings)
4. "Smart Street Lighting: A Guide to Implementing Intelligent Lighting Systems" (International Association of Lighting Management Companies)
5. "The Future of Street Lighting: Trends and Opportunities" (Navigant Research)
6. "Adaptive Street Lighting: A Whitepaper on the Benefits and Challenges of Smart Lighting" (Schreder)